COSC 50303: Data Mining
Course Project Description

Fall 2026

Purpose. The course project asks you to design and implement a research-driven data mining solution to a meaningful problem. You must define a clear question, prepare and understand the data, establish a credible baseline, evaluate suitable mining methods, and communicate the findings and limitations. The project is worth **30% of the final course grade**.

1 Project at a Glance

Component	Course Weight	Syllabus Timing
Team confirmation and project idea	5%	Team: week of August 31; proposal: week of September 14
Midterm progress report with implementation details	10%	Week of October 12, with progress presentations
Final report and implementation	10%	Exact deadline announced through TCU Online
Final presentation	5%	November 30, December 2, and December 7
Optional bonus credit	Up to 8%	Evaluated after the final presentations

The timing above follows the course syllabus. Submission links, exact due times, presentation assignments, and any approved schedule changes will be posted on TCU Online.

1.1 Team Rules

- **Undergraduate students:** Work individually or in a team of no more than three students.
- **Graduate students:** Complete the project individually.
- Team membership is fixed after confirmation unless the instructor approves a change.
- For an undergraduate team, one representative should send the confirmation with the subject `COSC-50303 Team [TeamName]` and copy all team members.
- All team members are responsible for understanding the complete project, not only the part they implemented.

2 Selecting an Appropriate Project

Choose one of the following project paths. Every topic requires instructor approval.

2.1 Path A: Predictive Data Mining

Develop and evaluate a solution for a supervised task such as classification, regression, or forecasting. The project should compare a simple baseline with one or more suitable data mining methods and explain how preprocessing, feature representation, validation, and metric selection affect the conclusions.

2.2 Path B: Pattern Discovery or Anomaly Detection

Study an unsupervised or descriptive problem using clustering, frequent-pattern mining, association analysis, anomaly detection, or related methods. The project must define how discovered structure will be validated. A collection of plots without an explicit mining question and evaluation plan is insufficient.

2.3 Path C: Mining Unconventional Data

Investigate time-series, spatial, spatio-temporal, graph, text, image, or another nontraditional data type. The method and validation scheme must respect the structure of the data. Examples include temporal holdouts for forecasting, spatial holdouts for geographic generalization, and sequence-aware similarity measures.

2.4 Path D: Reproduction, Extension, or Competition

Reproduce a clearly defined result from a paper or public solution and test a meaningful extension, or use an active competition as the problem and data source. Reproduction or a leaderboard score alone is insufficient. The project must include an independent baseline, controlled experiments, reproducible implementation, and analysis beyond the original source or public starter code.

Scope check. Data mining must be central to the project. A dashboard, exploratory analysis alone, or an off-the-shelf API call is not enough. The proposal must identify the mining question, comparison methods, evaluation evidence, and a feasible semester scope. Projects requiring unavailable data, excessive computation, or inaccessible proprietary services will not be approved.

2.5 Minimum Technical Expectations

Every project should include the following:

- a precise problem statement, inputs, outputs, and intended use of the result;
- a documented dataset with legal and ethical permission for the proposed use;
- an analysis of data types, quality, missingness, imbalance, and relevant preprocessing;
- appropriate similarity or dissimilarity measures when they affect the method;
- a defensible training, validation, and test protocol that avoids data leakage;
- at least one simple, credible baseline and one substantive data mining method;
- metrics or validation criteria appropriate to the task, data distribution, and project claims;
- a controlled comparison, sensitivity analysis, or ablation that isolates an important design choice;
- interpretation of representative successes, errors, clusters, patterns, or anomalies; and
- enough documentation to reproduce the main result.

Graduate projects should show greater independence and technical depth. This normally means a stronger literature foundation, a meaningful algorithmic or experimental extension, and a more systematic evaluation.

3 Deliverables

Required report format. Beginning with the project idea, all written project submissions must use the official IEEE Conference Template on Overleaf. Do not change the prescribed margins, font sizes, column widths, or line spacing. Submit the compiled PDF through TCU Online.

3.1 Team Confirmation and Project Idea (5%)

The project idea should be a concise one-page proposal, excluding references. It should answer five questions:

1. What data mining problem will you study, and why does it matter?
2. What dataset will you use, and can you access it now?
3. What data mining method will you test, and what will serve as the baseline?
4. How will you evaluate success while avoiding data leakage?
5. Why is the project feasible within the semester and available computing resources?

Include the project title, student or team name, all member names, dataset link, proposed evaluation metric, and two or more relevant references. Submit the proposal as a PDF through TCU Online. Approval may require a revision if the topic is too broad, too limited, or infeasible.

3.2 Midterm Progress Report with Implementation Details (10%)

The midterm report should be up to three pages, excluding references. It must show actual implementation progress rather than plans alone. Include:

- the updated problem statement and any change from the approved proposal;
- dataset statistics, data-quality findings, preprocessing, split strategy, and leakage checks;
- the baseline and proposed method, with enough algorithmic and implementation detail to understand the work;
- at least one completed experiment with a preliminary quantitative or otherwise defensible validation result;
- current difficulties, limitations, and a realistic plan for the remaining work; and
- a link to the working code or the requested code snapshot.

The report should use tables or figures when they communicate progress more clearly than prose. Every result must identify the data split, validation procedure, and metric or criterion being reported.

3.3 Final Report and Implementation (10%)

The final report must contain **exactly five pages of main text**, excluding references and optional appendices, and should read like a concise data mining paper. Appendices may provide supplementary evidence but may not contain material required to satisfy the grading criteria. Use the following structure when it fits the project:

1. **Abstract:** State the problem, approach, and main result in no more than 250 words.
2. **Introduction:** Explain the problem, motivation, contribution, and principal findings.

3. **Related Work:** Position the project relative to relevant published work and baselines.
4. **Data:** Describe provenance, attributes, size, quality, preprocessing, splits, and important limitations.
5. **Methods:** Explain the baseline, proposed mining methods, parameters, and major design choices.
6. **Experiments and Results:** Present the protocol, metrics, comparisons, and controlled analysis using readable tables or figures.
7. **Discussion and Conclusion:** Interpret the results, analyze errors or discovered patterns, state limitations, and identify a defensible next step.

The implementation must contain a clear README, dependency information, instructions for reproducing the main experiment, and the code needed to produce the reported result. Fix random seeds where practical and record the final configuration. Do not include large datasets or model files unless specifically requested; provide stable access instructions instead.

3.4 Final Presentation (5%)

The presentation should communicate the project as a coherent technical argument, not as a chronological list of tasks. The exact time limit and presentation assignment will be announced after the number of projects is finalized. **The instructor will provide the required slide template.**

A strong presentation should cover:

- the problem, motivation, and central data mining question;
- the data, preprocessing, and evaluation protocol;
- the baseline and proposed method, preferably with a clear pipeline figure;
- the most important quantitative or validation comparison;
- one useful error, cluster, pattern, anomaly, or case analysis;
- the main conclusion, limitations, and next step; and
- direct, technically accurate answers during questions.

All undergraduate team members must participate meaningfully and should be prepared to answer questions about any part of the project. Slides must be readable from the back of the classroom. Avoid dense paragraphs, raw code, and tables that are too small to read.

4 How the Work Will Be Evaluated

Area	Evidence Expected
Problem and scope	A precise, important, and feasible data mining question
Data understanding	Appropriate quality analysis, preprocessing, representation, and documentation
Technical quality	Suitable algorithms, correct implementation, and justified design decisions
Experimental quality	Sound validation, suitable metrics, credible baselines, controlled comparisons, and no leakage
Analysis	Honest interpretation, error or pattern analysis, limitations, and conclusions supported by evidence
Reproducibility	Organized code, recorded configuration, clear instructions, and traceable results
Communication	Clear writing, effective figures and tables, proper citations, and a focused presentation

Complexity alone does not earn credit. A simple method evaluated carefully is stronger than a sophisticated method used without a clear question or valid comparison.

5 Optional Bonus Credit (up to 8%)

The first four items below add **1.25 percentage points each** to the final course grade, for a maximum of **5 percentage points**. An eligible competition project may earn an additional **3 percentage points**, for an overall maximum of **8 percentage points**. All bonus work must be complete, clearly documented, and submitted with the final project.

1. **Actively maintained code repository:** Maintain a project repository with regular commits, including at least one substantive commit each week from every team member.
2. **Experiment tracking:** Use Weights & Biases (W&B) or a comparable platform such as MLflow, Comet, or ClearML to record configurations, metrics, plots, and experimental results. A free plan is sufficient; no paid service is required.
3. **Data or model interpretation:** Include and explain an appropriate interpretation technique, such as feature importance, SHAP values, partial-dependence analysis, cluster profiling, rule analysis, anomaly explanations, or an embedding visualization.
4. **Runnable demonstration:** Provide a simple, documented demonstration using a notebook, Gradio, Streamlit, or a comparable interface that accepts an example input and displays a meaningful result.
5. **Competition placement:** A project based on an instructor-approved active competition may earn an additional 3 percentage points if it finishes among the top five teams on the official final leaderboard. Evidence of the final placement is required.

6 Implementation, Attribution, and Academic Conduct

- You may use public papers, datasets, libraries, and implementations when their licenses permit it and when every source is cited clearly in both the report and code.
- Clearly distinguish code you wrote from code you adapted. Copying another student's or team's code is prohibited.
- AI tools may be used only when expressly permitted by the instructor. Any permitted use must follow

the instructor's conditions and be acknowledged. The evaluation, analysis, interpretation, synthesis, and final explanation must be your own work.

- You are responsible for verifying all generated text, code, citations, and results. Fabricated citations or unverified claims are academic and technical failures.
- Do not use private, sensitive, or restricted data without documented authorization. Medical, biometric, location, and human-subject data require particular care.
- Code must run. Before submission, start from a clean environment when possible and execute the complete evaluation path described in the README.

7 Finding a Topic or Dataset

Useful general starting points include UCI Machine Learning Repository, OpenML, Kaggle Datasets, Google Dataset Search, Harvard Dataverse, and Data.gov. Competition platforms include Kaggle, DrivenData, and Zindi.

Before committing to a dataset, verify its size, attribute and label quality, license, download access, compute requirements, privacy implications, and compatibility with the course timeline.

7.1 Popular Data Mining Dataset Collections and Benchmarks

The following resources may help you identify a suitable problem. Dataset licenses and access conditions must be checked before use.

- **UCI Machine Learning Repository:** Widely used datasets for classification, regression, clustering, and other tasks.
- **OpenML:** Versioned datasets, tasks, runs, and benchmark suites that support reproducible comparisons.
- **UCR Time Series Classification Archive:** A large benchmark collection for time-series classification.
- **FIMI Repository:** Transaction datasets used for frequent-itemset and association-rule mining.
- **Stanford SNAP:** Social, communication, citation, product, and other graph and network datasets.
- **MovieLens:** Rating datasets for recommendation and preference-mining experiments.
- **ODDS Library:** Benchmark datasets for outlier and anomaly detection.
- **NYC TLC Trip Records:** Large-scale taxi trip data suitable for temporal and spatial analysis.
- **USGS Earthquake Catalog:** Searchable event data for spatial and spatio-temporal mining.
- **NASA Earthdata:** Earth-observation products for environmental, spatial, and time-series projects.
- **Yelp Open Dataset:** Business, review, user, and interaction data for text, graph, and recommendation tasks.
- **GDELT Project:** Global news event and knowledge-graph data with geographic and temporal structure.

7.2 Examples of Active Competitions and Open Benchmarks

The following opportunities were open when this guide was prepared on September 11, 2026. Competition dates, eligibility, data access, and rules can change, so verify the official page before proposing one.

- **GEMS Prize Challenge** (ends December 3, 2026). Identify hidden geologic faults using spatial data; eligibility restrictions apply.
- **Richter's Predictor.** Multiclass classification using structured building and location data.
- **DengAI.** Time-series regression and disease forecasting using environmental data.
- **Pump it Up.** Classification of water-pump operating condition from structured data.
- **SNOMED CT Entity Linking.** Text mining in de-identified clinical notes; special access and licensing conditions apply.
- **Solar Filament Segmentation 2026.** Scientific-image segmentation and unconventional spatial data mining.

8 Sample Project Ideas

Dataset-size requirement. The project should use at least **100,000 usable instances** after preprocessing. For a substantially larger dataset, students may use a reproducible subset, but the resulting working dataset must still contain at least 100,000 instances. The proposal must clearly define what constitutes one instance.

8.1 Classification and Clustering

- **Forest Cover-Type Classification:** Predict one of seven forest cover types using cartographic and environmental attributes. Compare classification methods and evaluate performance under class imbalance. Use the UCI Covertype dataset (**581,012 instances**).
- **Forest-Region Cluster Discovery:** Cluster geographic observations using elevation, slope, hydrology, soil, and wilderness-area attributes. Compare the discovered clusters with known cover types without using the labels during clustering. Use the UCI Covertype dataset (**581,012 instances**).
- **Census Income Prediction and Subgroup Analysis:** Predict whether an individual's income exceeds a specified threshold and evaluate performance across demographic and employment groups. Use the UCI Census-Income (KDD) dataset (**299,285 instances**).
- **Rare Poker-Hand Classification:** Classify poker hands while addressing severe class imbalance and rare categories. Compare standard multiclass methods, resampling strategies, and cost-sensitive evaluation. Use the UCI Poker Hand dataset (**1,025,010 instances**).
- **Earthquake Building-Damage Prediction:** Predict the level of damage to buildings using structural and geographic attributes. Compare multiclass classification methods and analyze errors across building types and locations. Use the DrivenData Richter's Predictor dataset (**approximately 260,000 labeled buildings**).

8.2 Anomaly and Rare-Event Detection

- **Credit-Card Fraud Detection:** Detect fraudulent transactions under extreme class imbalance. Compare supervised classification with unsupervised or semi-supervised anomaly detection and evaluate precision, recall, PR-AUC, and alert burden. Use the Credit Card Fraud Detection dataset (**284,807 transactions**).
- **Network Intrusion Detection:** Identify normal traffic and multiple categories of malicious network activity. Compare classification and anomaly-detection methods and evaluate performance on rare attack categories. Use the UNSW-NB15 dataset (**approximately 2.54 million records**). A reproducible subset of at least 100,000 records may be used.
- **Household Electricity Anomaly Detection:** Detect unusual electricity-consumption periods while accounting for daily, weekly, and seasonal patterns. Use the UCI Individual Household Electric Power Consumption dataset (**approximately 2.08 million measurements**).

8.3 Time-Series Mining

- **Multi-Site Air-Quality Forecasting:** Predict pollutant concentrations using historical pollution measurements, meteorological variables, and monitoring-station information. Use temporal validation rather than a random split. Use the UCI Beijing Multi-Site Air Quality dataset (**420,768 hourly records**).

- **Air-Quality Pattern and Anomaly Analysis:** Cluster air-quality conditions across stations and identify unusual pollution events. Compare results across locations, seasons, and pollutants. Use the UCI Beijing Multi-Site Air Quality dataset (**420,768 hourly records**).
- **Household Electricity-Demand Forecasting:** Forecast future electricity consumption and compare simple statistical baselines with machine learning methods. Use the UCI Individual Household Electric Power Consumption dataset (**approximately 2.08 million measurements**).

8.4 Frequent-Pattern and Association Analysis

- **Web Clickstream Pattern Mining:** Apply Apriori, FP-Growth, or comparable methods to discover frequent navigation patterns and association rules from anonymized clickstream transactions. Use the FIMI Kosarak dataset (**990,002 transactions**).
- **Traffic-Accident Association Analysis:** Discover frequent combinations of road, vehicle, environmental, and accident conditions. Compare rules using support, confidence, lift, and redundancy measures. Use the FIMI Accidents dataset (**340,183 transactions**).
- **Online Retail Purchasing-Pattern Analysis:** Analyze products, cancellations, purchase times, countries, and transaction values. The project may investigate frequent patterns, unusual purchases, or product relationships. Use the UCI Online Retail dataset (**541,909 line-item records**).

8.5 Recommendation and Text Mining

- **Movie Recommendation:** Compare popularity, neighborhood-based collaborative filtering, and matrix factorization. Evaluate rating prediction and top-k recommendation quality. Use the MovieLens 1M dataset (**1,000,209 ratings**).
- **Large-Scale News Classification:** Classify news articles into topic categories and compare feature representations and classification algorithms. Use the AG News dataset (**127,600 articles across the standard training and test sets**).

8.6 Spatial and Spatio-Temporal Mining

- **Taxi-Demand Hotspot Discovery:** Apply spatial clustering to identify pickup and drop-off hotspots and examine how these patterns change by hour, weekday, and season. Use the NYC TLC Trip Record Data. Select a documented period containing at least **100,000 trips**.
- **Spatio-Temporal Taxi-Demand Prediction:** Divide the study area into geographic regions and predict future trip demand for each region. Use a chronological evaluation protocol and at least **100,000 trips** from the NYC TLC Trip Record Data.
- **Large-Scale Earthquake Pattern Mining:** Identify spatial clusters, temporal changes, and unusual seismic events. Construct a reproducible query containing at least **100,000 events** from the USGS Earthquake Catalog.

Important counting rules. For time-series projects, report the number of entities, sequences, time points, and constructed windows. For association-rule projects, count transactions or baskets rather than individual product rows. For spatial projects, define the geographic and temporal coverage. If sampling is required, use a documented random, stratified, temporal, or geographic procedure. Simply selecting the first 100,000 rows is generally unacceptable.